Alpha Institute — Technical Implementation Document

Based on the Actual Excel System (Alpha_Institute_ERP.xlsm)

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1. System Structure Overview

The system consists of **7 worksheets** that map directly to the modules the web application must implement:

#	Sheet (Arabic Name)	Purpose
1	لوحة التحكم (Dashboard)	KPIs and main statistics
2	الطلاب (Students)	Student records, hours, and billing
3	المعلمون (Teachers)	Teacher records, hours, and compensation
4	الحركات المالية (Financial Transactions)	All payments and expenses ledger
5	الإعدادات (Settings)	Prices, rates, and system constants
6	تقرير المعلم (Teacher Report)	Detailed per-teacher report
7	تقرير الطالب (Student Report)	Detailed per-student report

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2. Students Table

2.1 Fields and Columns

Col	Field Name	Data Type	Input/Calculated	Notes
A	Registration ID	Text	Auto	Auto-generated: `STD-0001`, `STD-0002`, ...
B	Full Name	Text	Input	Student’s full name
C	Phone	Number	Input	Student’s phone number
D	Guardian Phone	Number	Input	Parent/guardian phone number
E	Educational Stage	Text (Dropdown)	Input	Primary / Intermediate / Secondary / University / Foreign Schools / Special Needs
F	Grade	Number	Input	Grade number (1-12 or per stage)
G	Curriculum	Text (Dropdown)	Input	Ministry (Arabic), Ministry (English), etc.
H	Governorate	Text (Dropdown)	Input	Hawalli, Capital, Ahmadi, etc.
I	Area	Text	Input	Residential area

J	Address	Text	Input	Home address (block, street, house)
K	Sibling Group	Text	Input	Family group code (e.g., `FAM-01`)
L	Registration Date	Date	Input	Enrollment date
M	Subscription Hours (Added)	Number	Input	Hours purchased in this record
N	Total Amount (Added)	Number	Input	Total amount due for this record
O	Amount Paid (Added)	Number	Input	Amount actually paid for this record
P	Subject	Text (Dropdown)	Input	Math, Arabic, English, etc.
Q	Teacher	Text (Dropdown)	Input	Assigned teacher's name (linked to Teachers table)
R	Day 1	Text (Dropdown)	Input	First study day (Sunday, Monday, ...)
S	From (1)	Time	Input	Start time of day 1
T	To (1)	Time	Input	End time of day 1
U	Day 2	Text (Dropdown)	Input	Second study day (optional)
V	From (2)	Time	Input	Start time of day 2
W	To (2)	Time	Input	End time of day 2
X	Weekly Hours	Number	**Calculated**	Sum of (To − From) for both days
Y	Consumed Hours	Number	Input	Hours actually taught so far
Z	Total Balance	Number	**Calculated**	Sum of all subscription hours (col M) across all rows of same student
AA	Total Consumed	Number	**Calculated**	Sum of all consumed hours (col Y) across all rows of same student
AB	Total Due	Number	**Calculated**	Sum of all amounts (col N) across all rows of same student
AC	Total Paid	Number	**Calculated**	Sum of all payments (col O) across all rows of same

				student
AD	Remaining Hours	Number	**Calculated**	Total Balance − Total Consumed
AE	Remaining Amount	Number	**Calculated**	Total Due − Total Paid
AF	Payment Status	Text	**Calculated**	✅ Fully Paid / 🟡 Partially Paid / ❌ Not Paid / — No Dues
AG	Balance Alert	Text	**Calculated**	✅ OK / ⚠ Balance Running Low / ⚠ Hours Exceeded
AH	Teacher Due	Number	**Calculated**	Amount the teacher earns from this student’s record
AI	Primary Row	Boolean	**Calculated**	TRUE if this is the first row for this student (for deduplication in aggregates)
AJ	Notes	Text	Input	Free notes

2.2 Key Formulas (Business Logic)

Registration ID (A):

```
IF name is empty → empty
ELSE → "STD-" + row sequence padded to 4 digits
```

Weekly Hours (X):

```
Weekly Hours = (To1 − From1) in hours + (To2 − From2) in hours
Rounded to 1 decimal place.
If a day's times are missing, that day contributes 0.
```

Total Balance (Z): — a student can appear in multiple rows (renewals / multiple subjects)

```
Total Balance = SUM of Subscription Hours (M) for all rows where Name = this student
```

Total Consumed (AA):

```
Total Consumed = SUM of Consumed Hours (Y) for all rows where Name = this student
```

Total Due (AB):

```
Total Due = SUM of Total Amount (N) for all rows where Name = this student
```

Total Paid (AC):

```
Total Paid = SUM of Amount Paid (O) for all rows where Name = this student
```

Remaining Hours (AD):

```
Remaining Hours = Total Balance − Total Consumed
```

Remaining Amount (AE):

```
Remaining Amount = Total Due − Total Paid
```

Payment Status (AF):

```
IF Total Due = 0        → "— No Dues"
ELSE IF Remaining ≤ 0   → "✅ Fully Paid"
ELSE IF Total Paid > 0  → "🟡 Partially Paid"
ELSE                    → "❌ Not Paid"
```

Balance Alert (AG):

```
IF Remaining Hours < 0                          → "⚠ Hours Exceeded"
ELSE IF Remaining Hours ≤ LowHoursThreshold (2) → "⚠ Balance Running Low"
ELSE                                            → "✅ OK"
```

Teacher Due (AH): — the core compensation formula

```
Teacher Due = Consumed Hours (Y)
            × Hourly Rate for the student's Stage (from Settings)
            × Teacher Percentage (0.75 from Settings)
Rounded to 3 decimal places.
```

Primary Row (AI):

```
TRUE if this is the first occurrence of this student name in the table
(used so aggregate counts don't double-count students with multiple rows)
```

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3. Teachers Table

3.1 Fields and Columns

Col	Field Name	Data Type	Input/Calculated	Notes
A	Teacher ID	Text	Auto	Auto-generated: `TCH-0001`, …
B	Full Name	Text	Input	Teacher’s full name
C	Specialization	Text	Input	e.g., Class teacher, Math teacher
D	Phone	Number	Input	Teacher’s phone
E	Governorate	Text (Dropdown)	Input	Work area
F	Uses Institute Car	Boolean	Input	Yes / No
G	Status	Text (Dropdown)	Input	Active / Suspended / Retired
H	Subjects Taught	Text	Input	Comma-separated list
I	Registration Count	Number	**Calculated**	Count of student rows assigned to this teacher
J	Student Count	Number	**Calculated**	Count of UNIQUE students assigned to this teacher
K	Executed Hours	Number	**Calculated**	Total hours taught (sum of Consumed Hours of assigned students)
L	Due Before Deduction	Number	**Calculated**	Total earnings before car deduction
M	Transportation Deduction	Number	Input	Total car-usage deduction
N	Net Due	Number	**Calculated**	Due Before Deduction − Transportation Deduction
O	Paid to Teacher	Number	**Calculated**	Sum of “Teacher Payment” transactions for this teacher
P	Remaining Due	Number	**Calculated**	Net Due − Paid
Q	Notes	Text	Input	Free notes

3.2 Key Formulas

Registration Count (I):

```
COUNT of rows in Students table where Teacher = this teacher
```

Student Count (J):

```
COUNT of DISTINCT student names in Students table where Teacher = this teacher
```

Executed Hours (K):

```
SUM of Consumed Hours (Students.Y) where Students.Teacher = this teacher
Rounded to 1 decimal.
```

Due Before Deduction (L):

```
SUM of Teacher Due (Students.AH) where Students.Teacher = this teacher
Rounded to 3 decimals.
```

Net Due (N):

```
Net Due = Due Before Deduction (L) − Transportation Deduction (M, or 0 if empty)
```

Paid to Teacher (O):

```
SUM of Amount (F) in Financial Transactions
WHERE Type = "Teacher Payment" AND Name = this teacher
```

Remaining Due (P):

```
Remaining Due = Net Due (N) − Paid (O)
```

Important business rule: The transportation deduction is subtracted from the teacher's pay and **returns to the institute as revenue**. It is never double-counted.

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4. Financial Transactions Table (Ledger)

4.1 Fields and Columns

Col	Field Name	Data Type	Input/Calculated	Notes
A	Transaction ID	Text	Auto	Auto-generated: `MOV-0001`, …
B	Date	Date	Input	Transaction date
C	Type	Text (Dropdown)	Input	Student Payment / Teacher Payment / Expense
D	Student / Teacher	Text (Dropdown)	Input	Name (required for payments)
E	Item (for expenses)	Text	Input	Required only when Type = Expense (rent, electricity, …)
F	Amount	Number	Input	Transaction value (must be > 0)
G	Payment Method	Text (Dropdown)	Input	KNET / Bank Transfer / Cash / Cheque
H	Reference	Text	Input	Transaction/receipt reference number
I	Remaining (Student/Teacher)	Number	**Calculated**	Live remaining balance for the named person
J	Validation	Text	**Calculated**	OK / ⚠ Select a name / ⚠ Select an item / ⚠ Amount is zero
K	Notes	Text	Input	Free notes

4.2 Transaction Types

1. Student Payment (دفعة طالب) - Records money received from a student/family - Updates the student’s Total Paid → Remaining Amount → Payment Status automatically - Remaining column shows: student’s Remaining Amount after all payments

2. Teacher Payment (دفعة معلم) - Records money paid out to a teacher - Updates the teacher’s Paid → Remaining Due automatically - Remaining column shows: teacher’s Remaining Due after all payments

3. Expense (مصروف) - Institute operating expense (rent, fuel, supplies, …) - Requires the Item field - Included in profit/loss calculation

4.3 Validation Rules (must be enforced server-side)

```
IF Type is a payment AND Name is empty  → error "Select a name"
IF Type = Expense AND Item is empty     → error "Select an item"
IF Amount ≤ 0                           → error "Amount is zero"
ELSE                                    → OK
```

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5. Settings Table (System Constants)

5.1 General Settings

Setting	Value	Notes
Institute Name	Alpha Institute (معهد ألفا)	Shown in reports
Report Year	2026	Basis of monthly charts
Low Hours Alert Threshold	2	Alert when a student's remaining hours drop below this
Teacher Percentage	0.75 (75%)	**Fixed for all teachers and all stages** — the institute keeps 25%

5.2 Pricing Table (Hourly Rates by Stage)

Educational Stage	Hourly Rate (KD)
Primary (الابتدائية)	7
Intermediate (المتوسطة)	8
Secondary (الثانوية)	10
University (الجامعة)	10
Foreign Schools (المدارس الأجنبية)	15
Special Needs (الاحتياجات الخاصة)	10

> Note: In the Excel file the stored Primary value appears as 6.66667 due to an internal adjustment; the **canonical business price for Primary is 7 KD/hour**. In the web application, store the clean rates above as named constants in a Settings/Pricing table.

5.3 Other System Rules

Rule	Value	Applied To
Institute share of revenue	25%	Every taught hour
Car usage deduction	Fixed amount per session	Only teachers with "Uses Institute Car" = Yes
Sibling discount	10%	Second and subsequent students in the same family group

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6. Core Calculation Logic (Must Match Exactly)

6.1 Student Balance

```
Remaining Hours  = Purchased Hours (all records) − Consumed Hours (all records)
Remaining Amount = Total Due (all records) − Total Paid (all records)
```

6.2 Teacher Compensation

```
Gross Due (per student record) = Consumed Hours × Stage Hourly Rate × 0.75
Teacher Gross Total            = SUM of Gross Due across all assigned students
Net Due                        = Teacher Gross Total − Car Deduction Total
Remaining to Pay               = Net Due − Payments already made to teacher
```

Car deduction rules: - Applied per session when the teacher uses the institute's car - Subtracted once from the teacher's pay - Credited back to the institute (extra institute revenue — never double-subtracted)

6.3 Student Pricing

```
First student in family:
  Amount = Hours × Stage Hourly Rate

Second and subsequent students in same family group:
  Amount = Hours × Stage Hourly Rate × 0.90   (10% discount)
```

6.4 Institute Profit

```
Revenue        = SUM of all Student Payments
Teacher Cost   = SUM of all Net Teacher Dues
Expenses       = SUM of all Expense transactions
Car Recovery   = SUM of all car deductions (returns to institute)

Profit = Revenue − Teacher Cost − Expenses
(Car deductions are already inside Net Teacher Dues — do NOT subtract them again)
```

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7. Operational Workflows (Step by Step)

7.1 Adding a New Student

1. Fill personal data (Name, Phones, Stage, Grade, Curriculum, Governorate, Area, Address)
2. Assign Family Group code if the student has enrolled siblings
3. Select Subject and Teacher
4. Set study days and time slots (Day 1 / Day 2 with From–To times)
5. Enter Subscription Hours and the Total Amount (with sibling discount applied if applicable)
6. System auto-calculates: Registration ID, Weekly Hours, all balance columns, Payment Status, Balance Alert, Teacher Due

7.2 Recording a Session (Attendance)

1. Teacher completes the lesson at the student's home
2. Consumed Hours for that student record is increased by the session's hours
3. System automatically recalculates: Total Consumed, Remaining Hours, Balance Alert, Teacher Due
4. If the student was absent → hours are NOT consumed and the teacher earns nothing for that slot

7.3 Recording a Student Payment

1. Add a row in Financial Transactions: Type = "Student Payment", select student, enter Amount, Method, Reference
2. System automatically updates: Total Paid, Remaining Amount, Payment Status
3. The transaction row shows the student's live remaining balance

7.4 Recording Car Usage

1. Session is recorded normally
2. The per-session deduction is added to the teacher's Transportation Deduction total
3. Net Due decreases accordingly; the deduction amount is institute revenue

7.5 Monthly Teacher Settlement

1. System shows for each teacher: Executed Hours, Due Before Deduction, Car Deduction, Net Due, Already Paid, Remaining
2. Admin pays the Remaining amount
3. Payment is recorded as Type = "Teacher Payment" in Financial Transactions
4. Teacher's Remaining Due automatically returns to 0

7.6 Subscription Renewal

- A renewal = a **new row** for the same student name with new Subscription Hours and Amount
- All aggregate columns (Total Balance, Total Due, etc.) automatically include the new row because they sum across all rows of the same student

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8. Alerts and Status Indicators

Indicator	Condition	Display
Payment Status	Due = 0	— No Dues
Payment Status	Remaining ≤ 0	✅ Fully Paid
Payment Status	Paid > 0 and Remaining > 0	🟡 Partially Paid
Payment Status	Paid = 0 and Due > 0	❌ Not Paid
Balance Alert	Remaining Hours < 0	⚠ Hours Exceeded
Balance Alert	Remaining Hours ≤ 2	⚠ Balance Running Low
Balance Alert	Otherwise	✅ OK
Transaction Validation	Missing name/item or zero amount	⚠ with specific message

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9. Read-Only (Calculated) Fields Summary

These fields must **never be editable** in the UI — always computed server-side:

Students: Registration ID, Weekly Hours, Total Balance, Total Consumed, Total Due, Total Paid, Remaining Hours, Remaining Amount, Payment Status, Balance Alert, Teacher Due, Primary Row

Teachers: Teacher ID, Registration Count, Student Count, Executed Hours, Due Before Deduction, Net Due, Paid, Remaining Due

Transactions: Transaction ID, Remaining Balance, Validation Status

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10. Implementation Requirements for the Developer

1. **All calculated values are server-side only.** The frontend never sends computed values; the server never trusts them.
2. **Calculations must be idempotent** — recalculating always yields the same result.
3. **Use database transactions** for any linked operations (e.g., payment + balance update).
4. **Automatic recalculation** when any related record is added, edited, or deleted (equivalent to Excel's live formulas).
5. **All rates and percentages are named constants** in a Settings table — no hardcoded magic numbers.
6. **All numerals in UI and code are English digits (0–9)**, never Arabic-Indic (٩–٠).
7. **Dropdown-driven inputs** for: Stage, Curriculum, Governorate, Subject, Teacher, Days, Transaction Type, Payment Method, Status — matching the Excel data-validation lists.
8. **A student can have multiple records** (renewals / multiple subjects); aggregates always sum across all records of the same student.
9. **Audit logging** for every create/update/delete on financial data (who, when, what changed).
10. **Enforce the validation rules** from section 4.3 on every transaction before saving.

Document Date: July 17, 2026 **Version:** 2.0 — English, mirrors the actual Excel system **Status:** Ready for development